

Notes: Ke (JF, 2021)

Who Wears the Pants? Gender Identity Norms and Intrahousehold Financial Decision-Making

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1 Big picture

- **Question.** Does the gender of the financially sophisticated spouse matter for household financial choices, and can traditional gender norms explain why it matters?
- **Setting.** The paper studies household stock market participation in the United States using three large household surveys, and complements the observational evidence with a randomized online experiment.
- **Core challenge.** Household financial decisions are often made jointly, but the decision process inside the family is largely a black box. If a husband and a wife have similar financial sophistication, why should the household outcome differ depending on which spouse has that sophistication?
- **Main empirical result.** Households in which the husband works in finance are more likely to participate in the stock market than otherwise similar households in which the wife works in finance.
- **Main interpretation.** The paper argues that this gap is best explained by **gender identity norms**: even when the wife is the financially sophisticated spouse, her influence over intrahousehold financial decision-making is constrained.
- **Main experimental result.** In a randomized survey experiment, making female identity salient reduces women’s willingness to contribute correct financial advice to their husbands, but making male identity salient does not significantly affect men’s openness to correct advice from their wives.
- **Why this paper matters.** It moves household finance beyond the representative-household view and shows that household portfolio choices depend not only on preferences, wealth, and beliefs, but also on **who inside the household gets listened to**.

2 Data and measurement (key objects)

2.1 ASEC sample

- **Source.** Annual Social and Economic Supplement (ASEC) of the Current Population Survey, 1988–2018.
- **Sample restriction.** Nonmilitary married couples ages 24–64 who do not live in group quarters.
- **Unit of observation.** Household.
- **Financial outcome.** A dummy equal to 1 if any family member owns stock or mutual funds.
- **Main advantage.** Very large sample size, which lets the paper use a rich set of fixed effects. The ASEC also has a mini-panel feature, useful for within-household analysis, and a small overlap with the American Time Use Survey (ATUS), which is used to study social networks.
- **Sample size.** About 854,000 households in the main sample.

2.2 Census/ACS sample

- **Source.** The 5% samples of the 1980, 1990, and 2000 Censuses pooled with the American Community Survey, 2006–2015.
- **Sample restriction.** Again, nonmilitary married couples ages 24–64.
- **Financial outcome.** A dummy equal to 1 if the household reports investment income above \$500 or investment losses.
- **Main advantage.** Huge sample size and information on state of birth and ancestry, which the paper uses to proxy for traditional gender norms.
- **Sample size.** About 8.8 million households.

2.3 HRS sample

- **Source.** Health and Retirement Study, 1992–2016.
- **Sample restriction.** Married households ages 50–85.
- **Financial outcome.** A dummy equal to 1 if any household member owns stock or mutual funds.
- **Main advantage.** The HRS contains household wealth, as well as preferences and beliefs for *both* spouses, including risk tolerance, expectations, religious attendance, and a direct question on who has the final say in major family decisions.
- **Sample size.** About 52,000 household-year observations.

2.4 Experimental sample

- **Source.** A randomized controlled survey experiment run on Amazon Mechanical Turk.
- **Sample restriction.** Married U.S. residents ages 24–64.
- **Design.** Subjects first face an employee stock purchase plan (ESPP) decision, then complete a writing task that either primes gender identity or serves as a control, and finally face a spousal-disagreement scenario.
- **Main purpose.** To causally study whether gender identity changes (i) willingness to contribute correct financial ideas to one’s spouse and (ii) openness to correct advice from one’s spouse.
- **Scale.** Roughly 4,000 subjects overall, with the key contribution-stage regression using 1,891 female subjects and the aggregation-stage regression using 318 male subjects.

2.5 Key measured objects

- **Finance job.** A spouse works in finance if employed in depository institutions, nondepository credit institutions, security/commodity brokerage and investment companies, insurance, or real estate.
- **Husband in finance.** A dummy equal to 1 if the husband is the spouse working in finance.

- **Bank manager subsample.** Used to address the concern that men and women in finance may differ in rank or sophistication.
- **Risk preference.** From a hypothetical income-gamble question in the HRS.
- **Overconfidence.** Measured as self-rated memory percentile minus actual word-recall percentile.
- **Expectations.** Subjective percent chance that blue-chip stock mutual funds will be worth more in one year.
- **Social networks.** Measures of socializing/communicating and sports or recreational activities from the ATUS.
- **Gender-norm proxies.** Working-mother environment in birth state, ancestry linked to traditional plough agriculture, Southern upbringing, religious attendance, and reported household decision authority.

2.6 Unobservables and timing

- **Core unobservable concern.** The husband-finance versus wife-finance gap could be driven by omitted gender differences in rank, risk tolerance, overconfidence, beliefs, social networks, or broader unobserved household characteristics.
- **Timing.** The main observational evidence is cross-sectional or repeated cross-sectional. The career-switcher exercise uses short within-household changes over a one-year window, and the experiment provides truly randomized variation.
- **Interpretation challenge.** The paper is not trying to show that women know less. It is trying to show that, conditional on comparable financial sophistication, *women's influence is lower*.

3 Models and empirical specifications (all equations + interpretation)

This is a reduced-form paper, not a structural household model. The empirical specifications are designed to isolate whether household financial choices depend on *which spouse* is financially sophisticated.

3.1 Baseline regression

The core specification is

$$y_i = \alpha + \beta_1 \text{Finance}_i + \beta_2 \text{HusbandInFinance}_i + \gamma' X_i + \varepsilon_i,$$

where y_i is household stock market participation.

- **Interpretation of coefficients.**

$$\text{Effect of wife in finance} = \beta_1, \quad \text{Effect of husband in finance} = \beta_1 + \beta_2.$$

Therefore,

$$\text{husband-versus-wife gap} = \beta_2.$$

- **Controls.** The paper includes a rich set of fixed effects for the age, cohort, race, education, occupation, and self-employment status of both spouses; household income; relative income within the couple; homeownership; number of children; and state-year or region-year fixed effects. The HRS specifications also control for household wealth.
- **Why the specification is clever.** The paper compares households in which *one* spouse works in finance, and then asks whether it matters that the financially sophisticated spouse is the husband rather than the wife.
- **Main identifying comparison.** Two otherwise similar households each have one finance spouse, but in one household the finance spouse is the husband and in the other it is the wife.

3.2 Bank-manager specification

To address the concern that finance jobs may differ by rank across genders, the paper restricts attention to bank managers:

$$y_i = \alpha + \beta_1 \text{BankManager}_i + \beta_2 \text{HusbandBankManager}_i + \gamma' X_i + \varepsilon_i.$$

- **Interpretation.** This is the same logic as the baseline regression, but within a more homogeneous occupation group.
- **Why it matters.** If the baseline results were just about men occupying higher-level finance jobs, the husband-wife gap should weaken sharply within the bank-manager subsample. It does not.

3.3 Augmented regressions for alternative channels

The paper then augments the baseline regression with controls for risk preferences, overconfidence, expectations, and social networks:

$$y_i = \alpha + \beta_1 \text{Finance}_i + \beta_2 \text{HusbandInFinance}_i + \delta' Z_{h,i} + \theta' Z_{w,i} + \gamma' X_i + \varepsilon_i.$$

- **Examples of $Z_{h,i}$ and $Z_{w,i}$.** Husband and wife risk aversion, cognition, overconfidence, stock market expectations, and social-bonding activities.
- **Interpretation.** If the husband-wife finance gap disappears after these controls, then the main result may be driven by these gender differences rather than gender norms.
- **Result.** The coefficient on husband-in-finance remains very stable, so these alternative channels do not explain the main pattern.

3.4 Career-switcher specification

Using the ASEC mini-panel, the paper studies households in which neither spouse initially works in finance and the household initially does not hold stocks:

$$y_{i,t+1} = \alpha + \beta_1 \text{ToFinance}_{i,t+1} + \beta_2 \text{HusbandToFinance}_{i,t+1} + \delta \Delta \log(\text{HH income}_{i,t+1}) + \gamma' X_{i,t} + \varepsilon_{i,t+1}.$$

- **Effect of wife switching to finance.** β_1 .

- **Effect of husband switching to finance.** $\beta_1 + \beta_2$.
- **Gap.** β_2 .
- **Interpretation.** This design differences out many relatively fixed household traits, such as generalized trust, stock-market trust, cognitive ability, and other slowly changing household features.

3.5 Heterogeneity tests for gender norms

To test the gender-norm hypothesis directly, the paper interacts the baseline design with proxies for traditional attitudes:

$$y_i = \alpha + \beta_1 \text{Finance}_i + \beta_2 \text{HusbandInFinance}_i + \delta Z_i + \phi(Z_i \times \text{Finance}_i) + \psi(Z_i \times \text{HusbandInFinance}_i) + \gamma' X_i + \varepsilon_i.$$

- **Examples of Z_i .** Working-mother environment, traditional plough ancestry, Southern upbringing, active religious practice, and whether the husband has the final say.
- **Interpretation.** If gender norms are the mechanism, then the husband-wife finance gap should be larger exactly where traditional gender attitudes are stronger.
- **What the interaction means.** The interaction term with HusbandInFinance shows whether the incremental husband advantage is stronger in more traditional environments.

3.6 Experimental design

The experimental logic is built around an employee stock purchase plan (ESPP), which offers a discounted purchase and immediate resale opportunity.

- **Why ESPP is useful.** It creates a clean “correct” financial choice: participation is an arbitrage-like opportunity, so disagreement can be interpreted as failure to contribute or accept correct reasoning rather than a true difference in risk preferences.
- **Randomization.** Subjects are assigned either to
 - a **gender-identity prime**, in which they write about acting in a gender-congruent way in front of their spouse, or
 - a **control** writing task about resisting a default American lifestyle.
- **Two experimental arms.**
 - **Information contribution stage:** among subjects who would correctly choose the ESPP, test whether the subject is willing to contradict the spouse who is wrong.
 - **Information aggregation stage:** among subjects who initially make the wrong choice, test whether the subject accepts correct advice from the spouse.

3.7 Experimental regressions

For the information-contribution arm (female subjects who would themselves enroll in the ESPP), the paper estimates a probit model of the form

$$\mathbb{P}(\text{Contradict husband}_i = 1) = \Phi(\alpha + \beta \text{GenderIdentity}_i + \gamma' X_i).$$

For the information-aggregation arm (male subjects who initially decline the ESPP), the paper estimates

$$\mathbb{P}(\text{Follow wife's advice}_i = 1) = \Phi(\alpha + \beta \text{GenderIdentity}_i + \gamma' X_i).$$

- **Interpretation of β .** The causal effect of making gender identity salient.
- **Main result.** Female identity significantly reduces women's willingness to contribute correct advice, while male identity does not significantly change men's openness to correct advice.

4 Identification and instruments (what makes the estimates credible)

This is not an instrumental-variables paper in the standard sense. The identification comes from **stacking multiple research designs** that all point in the same direction.

4.1 What fails in a naive interpretation

- A simple cross-sectional difference between husband-finance and wife-finance households could reflect omitted heterogeneity in occupation rank, income, risk attitudes, beliefs, or social networks.
- It could also reflect selection: perhaps households in which wives work in finance are systematically different from those in which husbands do.
- The key question is therefore not whether the raw gap exists, but whether it survives after credible attempts to rule out alternative explanations.

4.2 Why the rich fixed-effects design helps

- The baseline regressions include detailed fixed effects for both spouses, not just one household head.
- This matters because recent changes in educational attainment and labor-market matching imply that households with financially sophisticated wives may differ in systematic ways from households with financially sophisticated husbands.
- Controlling for both spouses' occupations is especially important because it keeps the comparison within similar husband-wife occupation combinations.

4.3 Why the alternative-channel tests matter

- The HRS allows the author to control directly for risk aversion, overconfidence, and stock market expectations of both spouses.

- The ATUS merge allows the author to control for social-bonding activity.
- If the husband-wife finance gap were due to these channels, the coefficient on HusbandInFinance should shrink substantially. It does not.

4.4 Why the career-switcher design is informative

- The within-household design compares the same household before and after one spouse switches into finance.
- This strips out many slowly moving household attributes that are otherwise hard to observe.
- The fact that a husband's switch to finance raises participation more than a wife's switch is therefore much harder to explain by stable omitted heterogeneity.

4.5 Why the gender-norm heterogeneity results are informative

- The paper does not just show that a gap exists; it shows that the gap is **larger exactly where traditional gender role attitudes are stronger**.
- This pattern is observed across five very different proxies: working-mother exposure, plough ancestry, Southern culture, religiosity, and self-reported household authority.
- That repetition across unrelated sources of variation is the main reason the gender-norm interpretation is persuasive.

4.6 Why the experiment is especially strong

- The observational evidence can always be challenged on omitted-variable grounds.
- The experiment directly randomizes gender-identity salience, so it gives a clean causal test.
- The asymmetry is especially revealing: priming female identity reduces women's idea contribution, but priming male identity does not significantly change men's openness to advice. That is exactly the sort of mechanism the paper wants to isolate.

4.7 Relation to the literature

- Relative to standard household-finance work, the paper adds the family bargaining and identity dimension to portfolio participation.
- Relative to development and family economics, it studies intrahousehold power and norms in a rich-country, modern financial setting.
- Relative to identity economics, it shows that gender identity matters not just for preferences or confidence, but for who gets influence inside the household.

5 Tables and figures

5.1 Table I and Figure 1: Descriptive statistics and raw participation patterns

Table I
Descriptive Statistics

This table reports descriptive statistics for the three samples in this paper: the Annual Social and Economic Supplement (ASEC) of the Current Population Survey sample (1988–2018), the 5% sample of the decennial census (1980, 1990, and 2000) pooled with the American Community Survey (2006–2015), and the Health and Retirement Study (HRS) sample (1992–2016). The financial outcome variable for both the ASEC and the HRS samples is a dummy equal to 1 if the household owns any stock or mutual fund. The market participation variable for the Census sample is a dummy equal to 1 if the household reports investment income of greater than \$500 or investment losses. Both the ASEC and the Census samples are restricted to nonmilitary married couples aged between 24 and 64 who do not live in group quarters. The age range for the HRS sample is 50 through 85. Income and wealth variables are deflated in 2010 dollars by the price index for personal consumption expenditures.

	ASEC		Census		HRS	
	<i>N</i> = 854,013		<i>N</i> = 8,781,372		<i>N</i> = 52,238	
	Mean	<i>SD</i>	Mean	<i>SD</i>	Mean	<i>SD</i>
Market participation	0.30	0.46	0.20	0.40	0.36	0.48
Husband in finance	0.04	0.21	0.04	0.20	0.04	0.19
Wife in finance	0.06	0.34	0.06	0.24	0.06	0.23
Husband age	44.48	10.17	45.11	10.66	67.64	8.56
Wife age	42.32	9.98	42.90	10.53	64.60	8.55
Husband race						
White	0.74	0.44	0.80	0.40	0.80	0.40
Black	0.07	0.25	0.06	0.24	0.10	0.30
Other	0.19	0.40	0.14	0.34	0.10	0.30
Wife race						
White	0.73	0.44	0.80	0.40	0.80	0.40
Black	0.06	0.24	0.06	0.23	0.10	0.30
Other	0.20	0.40	0.15	0.35	0.10	0.30
Husband education						
Less than high school	0.16	0.37	0.13	0.34	0.23	0.42
High school graduate	0.26	0.44	0.34	0.47	0.33	0.47
Some college	0.25	0.43	0.22	0.42	0.19	0.39
College graduate	0.33	0.47	0.31	0.46	0.25	0.43
Wife education						
Less than high school	0.15	0.36	0.11	0.31	0.19	0.39
High school graduate	0.26	0.44	0.37	0.48	0.41	0.49
Some college	0.27	0.44	0.24	0.43	0.23	0.42
College graduate	0.32	0.47	0.28	0.45	0.18	0.39
Husband self-employed	0.14	0.35	0.14	0.34	0.13	0.33
Wife self-employed	0.07	0.25	0.07	0.25	0.06	0.25
Log husband income	10.66	1.03	10.71	0.89	10.22	0.88
Log wife income	9.52	1.92	9.86	1.23	9.49	1.02
Log household income	11.17	0.79	11.17	0.75	10.89	0.86
Log household wealth					12.31	1.54
Home ownership	0.81	0.39	0.82	0.38	0.93	0.26
Number of children	1.44	1.25	1.32	1.26	2.86	1.81

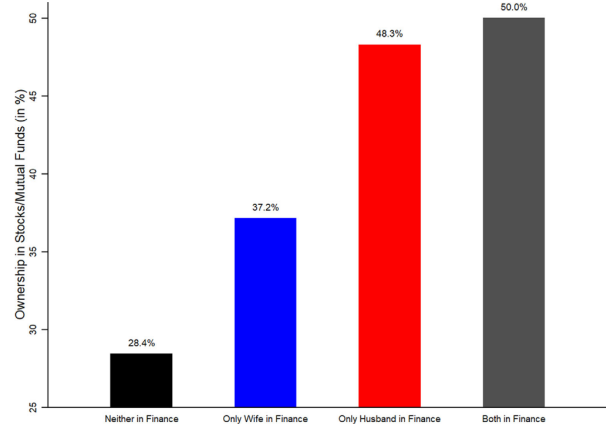


Figure 1. Career in finance and household stock market participation. This figure summarizes stock market participation rates among households in which neither spouse works in finance, those in which only the wife works in finance, those in which only the husband works in finance, and those in which both spouses work in finance. A household participates in the stock market if any family member holds any stock or mutual fund. Data are from the Annual Social and Economic Supplement of the Current Population Survey for 1988–2018 and the sample is limited to nonmilitary married couples aged between 24 and 64 who do not live in group quarters. (Color figure can be viewed at wileyonlinelibrary.com)

Read:

- The raw data already point to the main result. In the ASEC sample, stock market participation is about 28% when neither spouse works in finance, 37% when only the wife works in finance, and 48% when only the husband works in finance.
- The descriptive statistics also show why multiple samples are useful: the ASEC gives a very large modern cross section, the Census gives ancestry and birthplace, and the HRS gives wealth and spousal beliefs.
- A potential concern is that more women than men work in finance in the raw data because of administrative jobs. The paper anticipates this and later shows the result survives occupation controls and finance-job reclassification.

5.2 Table II: Baseline regressions

Read:

- This is the core result. In the ASEC sample, a wife in finance raises participation by about 3 percentage points, while a husband in finance raises it by about 5 percentage points, implying a husband-versus-wife gap of 2 percentage points.

- In the Census sample, the incremental gap is about 1.6 percentage points.
- In the HRS sample, the incremental gap is about 6.7–6.9 percentage points, even after controlling for household wealth.
- The paper interprets these as economically meaningful gaps, roughly 5% to 16% of the participation rate among the wife-finance households depending on sample.

Table II
Baseline Regressions

This table reports the baseline regression results. The dependent variable for both the ASEC and the HRS samples is a dummy equal to 1 if the household owns any stock or mutual fund. The dependent variable for the Census sample is a dummy equal to 1 if the household reports investment income of greater than \$500 or investment losses. Husband in finance indicates whether the husband works in finance. Finance indicates whether one spouse works in finance. Age is grouped into three-year intervals and birth cohorts are defined as 10-year birth intervals. Household income is classified into 28 bins: $(n-1) \times \$10,000 \leq \text{household income} < n \times \$10,000$ ($n = 1, 2, \dots, 20$), $\$150,000 + n \times \$50,000 \leq \text{household income} < \$200,000 + n \times \$50,000$ ($n = 1, 2, \dots, 6$), household income $\geq \$500,000$, and negative household income. Relative income is the ratio of the husband's income to the wife's income and is classified into 23 bins: $n < \text{relative income} \leq n+1$ ($n = 1, 2, \dots, 9$), $1/(n+1) \leq \text{relative income} < 1/n$ ($n = 1, 2, \dots, 9$), relative income > 10 , relative income < 0.1 , nonpositive husband's income and positive wife's income, positive husband's income and nonpositive wife's income, and nonpositive income from both spouses. Occupations are grouped into 11 broad categories following Acemoglu and Autor (2011). Standard errors in parentheses are clustered at the state level for both the ASEC and the Census samples. Robust standard errors are reported for the HRS sample. Levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	ASEC	Census	HRS	
	(1)	(2)	(3)	(4)
Husband in finance	0.020*** (0.004)	0.016*** (0.001)	0.069*** (0.014)	0.067*** (0.013)
Finance	0.030*** (0.002)	0.007*** (0.001)	0.016* (0.009)	0.009 (0.009)
Log household wealth				0.110*** (0.002)
<i>Fixed Effects:</i>				
Husband age group	Yes	Yes	Yes	Yes
Wife age group	Yes	Yes	Yes	Yes
Husband cohort of birth	Yes	Yes	Yes	Yes
Wife cohort of birth	Yes	Yes	Yes	Yes
Husband race $\times$ Wife race	Yes	Yes	Yes	Yes
Husband edu. $\times$ Wife edu.	Yes	Yes	Yes	Yes
Husband occ. $\times$ Wife occ.	Yes	Yes	Yes	Yes
Husband self-employed	Yes	Yes	Yes	Yes
Wife self-employed	Yes	Yes	Yes	Yes
Household income	Yes	Yes	Yes	Yes
Relative income	Yes	Yes	Yes	Yes
Home ownership	Yes	Yes	Yes	Yes
Number of children	Yes	Yes	Yes	Yes
State $\times$ Year	Yes	Yes	No	No
Region $\times$ Year	No	No	Yes	Yes
Observations	854,013	8,781,372	52,238	52,238
Adj. R^2	0.220	0.198	0.207	0.265

Table III
Evidence from Bank Managers

This table reports OLS estimates of the gap in stock market participation between households in which the husband works as a bank manager and those in which the wife works as a bank manager. The dependent variable for the ASEC sample is a dummy equal to 1 if the household owns any stock or mutual fund. The dependent variable for the Census sample is a dummy equal to 1 if the household reports investment income of greater than \$500 or investment losses. Husband bank manager indicates whether the husband works as a bank manager. Bank manager indicates whether one spouse works as a bank manager. All of the fixed effects in Table II are included as controls except for the occupation fixed effects. Standard errors in parentheses are clustered at the state level and levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	ASEC (1)	Census (2)
Husband bank manager	0.055*** (0.018)	0.035*** (0.004)
Bank manager	0.087*** (0.010)	0.022*** (0.003)
Controls	Yes	Yes
Observations	854,013	8,781,372
Adj. R^2	0.215	0.194

5.3 Table III: Evidence from bank managers

Read:

- The bank-manager subsample addresses the claim that men in finance simply occupy more senior jobs.
- In the ASEC sample, a wife bank manager raises participation by about 8.7 percentage points, while a husband bank manager raises it by about 14.2 percentage points, leaving a gap of 5.5 percentage points.
- In the Census sample, the implied gap is about 3.5 percentage points.
- So the result is not driven just by broad occupation categories hiding within-finance rank differences.

Table IV
The Effect of Risk Preferences

This table relies on the HRS sample and analyzes the effect of risk preferences. The dependent variable is a dummy equal to 1 if the household owns any stock or mutual fund. Husband in finance indicates whether the husband works in finance. Finance indicates whether one spouse works in finance. The risk-aversion measures are based on a hypothetical income gamble question, namely, whether the respondent would take a job (i) with equal probability of doubling income or cutting it in half, (ii) with equal probability of doubling income or cutting it by a third, (iii) with equal probability of doubling income or cutting it by 20%, or (iv) none of the above. The sample is restricted to households in which both spouses answer the income gamble question. All of the fixed effects in Table II and log household wealth are included as controls. Robust standard errors are reported in parentheses and levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Husband in finance	0.068*** (0.015)	0.066*** (0.015)	0.068*** (0.015)	0.066*** (0.015)
Finance	0.007 (0.010)	0.007 (0.010)	0.007 (0.010)	0.007 (0.010)
Husband risk aversion		-0.008*** (0.002)		-0.008*** (0.002)
Wife risk aversion			-0.003 (0.002)	-0.002 (0.002)
Controls	Yes	Yes	Yes	Yes
Observations	39,987	39,987	39,987	39,987
Adj. R^2	0.266	0.266	0.266	0.266

Table V
The Effect of Overconfidence

This table relies on the HRS sample and analyzes the effect of overconfidence. The dependent variable is a dummy equal to 1 if the household owns any stock or mutual fund. Husband in finance indicates whether the husband works in finance. Finance indicates whether one spouse works in finance. The cognition measures are based on word recall tasks, with scores ranging from 0 and 20. Overconfidence is defined as the difference between self-rated memory capability and actual performance on word recall tasks, both in percentile ranks. The sample is restricted to households in which both spouses self-rate their memory capability and complete the word recall task. All of the fixed effects in Table II and log household wealth are included as controls. Robust standard errors are reported in parentheses and levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Husband in finance	0.066*** (0.014)	0.065*** (0.014)	0.066*** (0.014)	0.065*** (0.014)
Finance	0.008 (0.009)	0.007 (0.009)	0.008 (0.009)	0.008 (0.009)
Husband cognition		0.011*** (0.001)		0.011*** (0.001)
Husband overconfidence		0.042*** (0.007)		0.040*** (0.007)
Wife cognition			0.005*** (0.001)	0.003** (0.001)
Wife overconfidence			0.017** (0.008)	0.011 (0.008)
Controls	Yes	Yes	Yes	Yes
Observations	50,449	50,449	50,449	50,449
Adj. R^2	0.265	0.266	0.265	0.266

Table VI
The Effect of Expectations

This table relies on the HRS sample and analyzes the effect of expectations. The dependent variable is a dummy equal to 1 if the household owns any stock or mutual fund. Husband in finance indicates whether the husband works in finance. Finance indicates whether one spouse works in finance. The expectation measures are the percent likelihood that the respondent believes that mutual fund shares invested in blue chip stocks like those in the Dow Jones Industrial Average will be worth more than they are today by the same time next year. The sample is restricted to households in which both spouses make their predictions. All of the fixed effects in Table II and log household wealth are included as controls. Robust standard errors are reported in parentheses and levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Husband in finance	0.067*** (0.023)	0.064*** (0.023)	0.068*** (0.023)	0.065*** (0.023)
Finance	-0.018 (0.015)	-0.017 (0.015)	-0.019 (0.015)	-0.018 (0.015)
Husband expectation		0.096*** (0.013)		0.088*** (0.013)
Wife expectation			0.076*** (0.014)	0.063*** (0.014)
Controls	Yes	Yes	Yes	Yes
Observations	15,443	15,443	15,443	15,443
Adj. R^2	0.254	0.256	0.255	0.257

5.4 Tables IV–VI: Risk preferences, overconfidence, and expectations

Read:

- These tables test the most natural omitted-variable stories from household finance.
- Controlling for both spouses’ risk preferences barely changes the husband-finance coefficient.
- The same is true when controlling for cognition and overconfidence, and when controlling for both spouses’ expectations of future stock returns.
- The message is that the main gap is not just “men are less risk averse,” “men are more overconfident,” or “men are more optimistic.”

5.5 Table VII: Social networks

Read:

- One alternative story is that men in finance may gather more investment-relevant information through social interactions.

- Social bonding itself matters for participation, but adding it as a control leaves the husband-wife finance gap almost unchanged.
- In the ATUS-linked sample, the gap remains around 6 percentage points.
- So gender differences in social networking are not enough to explain the main result.

Table VII
The Effect of Social Networks

This table relies on the ASEC sample merged with the American Time Use Survey (ATUS) and analyzes the effect of social networks. The dependent variable is a dummy equal to 1 if the household owns any stock or mutual fund. Husband in finance indicates whether the husband works in finance. Finance indicates whether one spouse works in finance. Social bonding indicates whether the ATUS respondent spent any time the previous day (i) socializing and communicating (e.g., talking with neighbors, spending time with friends, attending receptions, and going to parties) in column (2), (ii) attending or participating in sports or recreational events (e.g., playing tennis, golfing, and watching a football game) in column (3), and (iii) doing either (i) or (ii) in column (4). I further require that the spouse who works in finance be the ATUS respondent. All of the fixed effects in Table II are included as controls. Standard errors in parentheses are clustered at the state level and levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Husband in finance	0.060*** (0.020)	0.061*** (0.020)	0.059*** (0.020)	0.061*** (0.020)
Finance	0.018* (0.010)	0.017* (0.010)	0.018* (0.010)	0.017* (0.010)
Social bonding		0.018*** (0.006)	0.023*** (0.006)	0.022*** (0.005)
Controls	Yes	Yes	Yes	Yes
Observations	41,862	41,862	41,862	41,862
Adj. R^2	0.235	0.236	0.236	0.236

Table VIII
Evidence from Career Switchers

This table relies on the ASEC sample limited to married couples who are interviewed twice in two consecutive years. I further require that neither spouse worked in finance and that the household did not invest in any stock or mutual fund in the first year. The unit of analysis is the household in the second year and the dependent variable is a dummy equal to 1 if the household owns any stock or mutual fund. Husband to finance indicates whether the husband switches his career to finance. To finance indicates whether one spouse switches to finance. Δ log household income is the change in log household income. Husband's occupation by wife's occupation fixed effects are included in columns (2) to (4). All of the fixed effects in Table II for the previous year are included as controls in column (4). Standard errors in parentheses are clustered at the state level and levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	(1)	(2)	(3)	(4)
Husband to finance	0.045*** (0.014)	0.033** (0.014)	0.033** (0.014)	0.028** (0.013)
To finance	0.091*** (0.009)	0.061*** (0.009)	0.059*** (0.009)	0.061*** (0.009)
Δ log household income			0.031*** (0.002)	0.065*** (0.002)
Husband occ. $\times$ Wife occ.	No	Yes	Yes	Yes
Controls	No	No	No	Yes
Observations	134,906	134,906	134,906	134,906
Adj. R^2	0.002	0.044	0.047	0.100

5.6 Table VIII: Evidence from career switchers

Read:

- This is the strongest nonexperimental design in the paper.
- In the richest specification, when a wife switches into finance, household participation rises by about 6.1 percentage points; when a husband switches into finance, it rises by about 8.9 percentage points.
- The implied within-household gap is therefore about 2.8 percentage points even after controlling for changes in household income and prior household characteristics.
- This makes it much harder to argue that the baseline result is entirely driven by fixed household differences.

5.7 Table IX: Working mothers

Read:

- Exposure to working mothers is used as a proxy for less traditional gender-role attitudes.
- Among households more likely to have been exposed to working mothers, the finance-spouse gap is about 1.1 percentage points; among households less exposed, it is about 2.6 percentage points.
- The interaction terms show that working-mother exposure dampens the husband advantage.
- This is exactly what the gender-norm mechanism predicts.

Table IX
The Influence of Working Mothers

This table relies on the Census subsample of white natives and analyzes the influence of working mothers. The dependent variable is a dummy equal to 1 if the household reports investment income of greater than \$500 or investment losses. Employment ratio is the ratio of the number of working to nonworking women aged between 30 and 35 in the birth state of the husband (wife) when he (she) was born. Working mother is a dummy equal to 1 if the employment ratio is above the sample median. Husband in finance indicates whether the husband works in finance. Finance indicates whether one spouse works in finance. All of the fixed effects in Table II and the spouse's cohort of birth by state of birth fixed effects are included as controls. Standard errors in parentheses are clustered at the state level and levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	Husband's Side (1)	Wife's Side (2)
Working mother $\times$ Husband in finance	-0.015*** (0.003)	-0.016*** (0.003)
Working mother $\times$ Finance	-0.011*** (0.001)	-0.011*** (0.001)
Working mother	0.003*** (0.001)	0.001 (0.001)
Husband in finance	0.026*** (0.002)	0.026*** (0.002)
Finance	0.012*** (0.001)	0.012*** (0.001)
Controls	Yes	Yes
Observations	6,240,888	6,240,888
Adj. R^2	0.199	0.199

5.8 Table X: Traditional gender norms across cultural environments

Read:

- **Panel A (traditional plough ancestry):** the husband-wife finance gap is much stronger among descendants of societies with historically more traditional gender roles.
- **Panel B (Southern culture):** the gap is larger in Southern families, especially from the husband's side.
- **Panel C (religion):** among households with a nonreligious husband, the gap is about 3.5 percentage points; among those with an actively religious husband, it is about 11.6 percentage points. A similar amplification appears using the wife's religiosity.
- **Panel D (who wears the pants):** when the husband is reported to have the final say, the participation gap is about 12.1 percentage points; when he does not, the gap is about 6.0 percentage points. This is the paper's most direct observational evidence on intrahousehold authority.

5.9 Figure 2: Experimental design

[?] Read:

- The experiment has a clear two-stage logic: first establish a correct or incorrect financial choice with the ESPP, then randomize gender-identity salience, then introduce disagreement with the spouse.
- The design separates **information contribution** from **information aggregation**, which is important because norms could operate through either stage.
- This figure is what makes the experiment easy to understand: the treatment changes identity salience, not financial sophistication.

Table X
Traditional Gender Norms across Cultural Environments

This table analyzes the role of traditional gender norms across cultural environments. Panels A and B rely on the Census sample, and Panels C and D rely on the HRS sample. The dependent variable is a dummy equal to 1 if the household reports investment income of greater than \$500 or investment losses in Panels A and B and is a dummy equal to 1 if the household owns any stock or mutual fund in Panels C and D. In Panel A, the sample is limited to natives who speak foreign languages at home with foreign ancestries. In Panel B, the sample is limited to white natives residing in the state in which one spouse was born. Traditional plough use indicates whether the fraction of citizens with ancestors that used plough cultivation in the reported country of ancestry exceeds 50%. Southern family indicates whether state of birth is in the south. Actively religious indicates whether the frequency of attending religious services is at least once a week. Husband final say indicates whether the husband has the final say when it comes to major family decisions. Husband in finance indicates whether the husband works in finance. Finance indicates whether one spouse works in finance. All of the fixed effects in Table II are included as controls. The spouse's ancestry and state of birth fixed effects are additionally included in Panels A and B, respectively, and log household wealth is controlled for in Panels C and D. Standard errors in parentheses are clustered at the state level in Panels A and B, and robust standard errors are reported in Panels C and D. Levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

Panel A: Traditional Plough Practice		
	Husband's Side (1)	Wife's Side (2)
Traditional plough use × Husband in finance	0.050*** (0.007)	0.034*** (0.008)
Traditional plough use × Finance	−0.005 (0.007)	−0.004 (0.005)
Traditional plough use	0.016*** (0.003)	0.009 (0.005)
Husband in finance	−0.018*** (0.006)	−0.009 (0.007)
Finance	0.016*** (0.005)	0.017*** (0.003)
Controls	Yes	Yes
Observations	220,051	238,609
Adj. R^2	0.201	0.203

Panel B: Southern Culture		
	Husband's Side (1)	Wife's Side (2)
Southern family × Husband in finance	0.011*** (0.004)	0.006 (0.004)
Southern family × Finance	0.001 (0.002)	0.000 (0.003)
Husband in finance	0.017*** (0.003)	0.016*** (0.003)
Finance	0.007*** (0.001)	0.008*** (0.001)
Controls	Yes	Yes
Observations	3,933,621	3,918,871
Adj. R^2	0.185	0.184

(Continued)

Table X—Continued

Panel C: Religion		
	Husband's Side (1)	Wife's Side (2)
Actively religious × Husband in finance	0.082*** (0.026)	0.044* (0.026)
Actively religious × Finance	−0.047*** (0.018)	−0.050*** (0.017)
Actively religious	0.004 (0.004)	0.009** (0.004)
Husband in finance	0.035** (0.017)	0.048*** (0.018)
Finance	0.027** (0.011)	0.032*** (0.012)
Controls	Yes	Yes
Observations	51,271	51,688
Adj. R^2	0.265	0.265

Panel D: Who Wears the Pants		
	Husband's Side (1)	Wife's Side (2)
Husband final say × Husband in finance	0.061* (0.032)	0.032 (0.032)
Husband final say × Finance	−0.039* (0.023)	−0.028 (0.022)
Husband final say	−0.009* (0.005)	0.009* (0.005)
Husband in finance	0.060*** (0.017)	0.067*** (0.017)
Finance	0.008 (0.011)	0.007 (0.012)
Controls	Yes	Yes
Observations	40,191	40,191
Adj. R^2	0.267	0.267

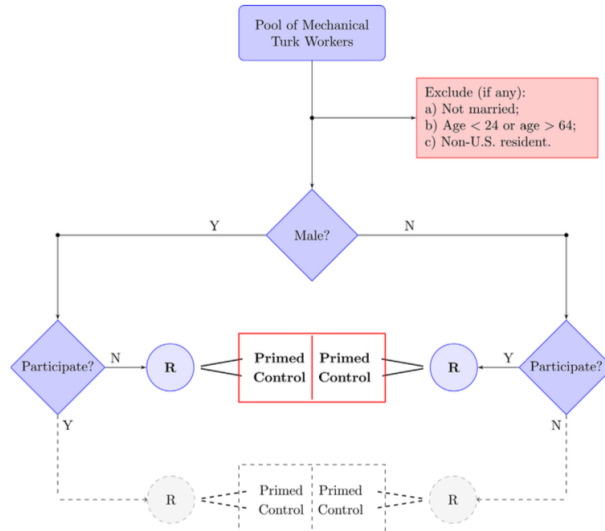


Figure 2. Experimental design. This figure summarizes the design of the randomized controlled experiment. Subjects are recruited from Amazon's Mechanical Turk and the sample is restricted to married individuals aged between 24 and 64 who are U.S. residents. They are first introduced to an employee stock purchase plan and are asked whether they plan to participate. All subjects then complete a writing task in which some of them are randomly assigned to the primed condition where gender identity is made salient to them, whereas the rest are assigned to the control condition. Finally, subjects are asked to make the participation decision again in a designed scenario in which their spouse has an opposing viewpoint. (Color figure can be viewed at wileyonlinelibrary.com)

5.10 Figure 3 and Table XI: Causal impacts of gender identity

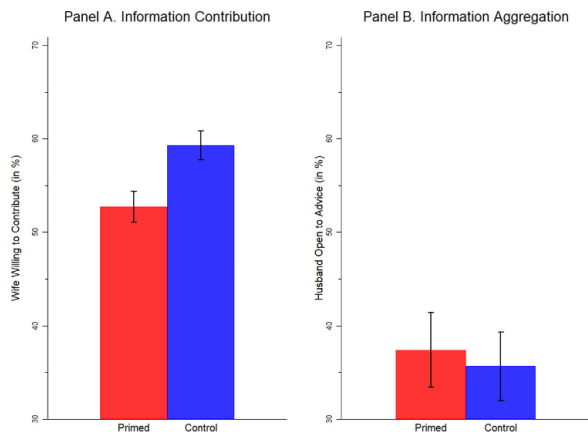


Figure 3. Causal impacts of gender identity. This figure presents univariate evidence from a randomized controlled experiment designed to study the causal impacts of gender identity on intrahousehold financial decision-making. Panel A plots, by randomization condition, women's influence at the information contribution stage of the financial decision-making process between spouses. The sample is restricted to female subjects who chose to participate in the employee stock purchase plan. The subject is willing to contribute ideas to her husband if she contradicts her husband, who is an eligible employee but not inclined to enroll in the plan. Panel B plots, by randomization condition, men's influence at the information aggregation stage of the financial decision-making process between spouses. The sample is restricted to male subjects who did not choose to enroll in the plan. The subject is open to advice from his wife if he chooses to participate in the plan after hearing the advice from his wife, who by design has the correct reasoning. In both panels, error bars indicate standard errors of the arithmetic mean. (Color figure can be viewed at wileyonlinelibrary.com)

Table XI
Causal Impacts of Gender Identity

This table reports causal impacts of gender identity on intrahousehold financial decision-making. The first two columns report results for the information contribution stage. The sample is restricted to female subjects who chose to participate in the employee stock purchase plan. The dependent variable of the probit regression is a dummy equal to 1 if the wife contradicts her husband, who is an eligible employee but not inclined to enroll in the plan. The next two columns report results for the information aggregation stage. The sample is restricted to male subjects who did not choose to participate in the plan. The dependent variable is a dummy equal to 1 if the husband follows the advice from his wife, who by design has the correct reasoning. Gender identity is a dummy equal to 1 if the subject is primed with gender identity, and all of the control variables are defined in Table IA.VIII in the Internet Appendix. Coefficients are reported as marginal effects and robust standard errors are reported in parentheses. Levels of significance are denoted as follows: * $p < 0.10$, ** $p < 0.05$, and *** $p < 0.01$.

	Information Contribution Stage		Information Aggregation Stage	
	(1)	(2)	(3)	(4)
Gender identity	-0.066*** (0.023)	-0.070*** (0.023)	0.017 (0.054)	0.026 (0.056)
Age		0.004*** (0.001)		-0.002 (0.003)
White		0.026 (0.032)		-0.008 (0.073)
College graduate		-0.052** (0.026)		0.094 (0.063)
Full-time employed		0.067*** (0.024)		0.042 (0.071)
Log family income		-0.007 (0.020)		0.022 (0.047)
Having child		-0.016 (0.027)		-0.073 (0.060)
Trading experience		0.105*** (0.027)		-0.109 (0.073)
Financial literacy		0.118*** (0.025)		-0.032 (0.061)
Risk aversion		-0.154*** (0.025)		0.013 (0.062)
Generalized trust		0.018 (0.024)		-0.036 (0.056)
Observations	1,891	1,891	318	318
Pseudo- R^2	0.003	0.062	0.000	0.022

Read:

- **Panel A / contribution stage:** among women who would make the correct ESPP decision themselves, 59% in the control group contradict their husband, versus 53% in the gender-identity prime.
- The probit estimates imply that female identity reduces women's willingness to contribute correct advice by about 7 percentage points.
- **Panel B / aggregation stage:** among men who initially make the wrong decision, about 37% in the primed group and 36% in the control group follow correct advice from their wives; the treatment effect is statistically insignificant.
- The main causal mechanism is therefore not that men become less receptive when male identity is salient. It is that women become less likely to push correct ideas when female identity is salient.

6 Short summary

- **Conclusion.** Households are more likely to participate in the stock market when the financially sophisticated spouse is the husband rather than the wife.
- **Identification insight.** The paper triangulates this fact using rich observational controls, within-household career switches, heterogeneity by gender norms, and a randomized experiment.
- **Main mechanism.** Traditional gender identity norms constrain the wife's influence over intrahousehold financial decision-making, especially at the stage of contributing ideas.

7 Conclusion

7.1 Core conclusion (what the paper establishes)

- The paper establishes that intrahousehold financial outcomes depend not only on whether a household has financial sophistication, but also on *which spouse* has it.
- A wife's financial sophistication translates less strongly into household stock market participation than an equally sophisticated husband's.
- This gap is not well explained by differences in risk tolerance, overconfidence, expectations, social networks, or time-invariant household heterogeneity.

7.2 Economic intuition

- In a frictionless household with equal influence, it should not matter whether the husband or the wife is the finance professional.
- But if traditional norms assign more authority to the husband, then the wife's financial knowledge is not converted into household action as effectively.
- The experiment clarifies one concrete mechanism: identity norms reduce women's willingness to press correct ideas, even when they know the right answer.

7.3 Takeaway

This paper is best read as a household-finance paper about **power**, not just participation. Its main contribution is to show that portfolio choice inside families is shaped by social norms governing authority and voice. The empirical result is simple but powerful: a financially sophisticated wife does not shift household stock market participation as much as a financially sophisticated husband. The broader lesson is that financial capability inside the household is not enough; what also matters is whether that capability has influence.